

Preference-Conditioned Capability: From B-IQ to Goal-Realizing LLM Agent Loops

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Abstract

Large-language models (LLMs) excel at raw prediction, but deployed agent loops require competence that is conditioned on *what the agent wants*. Building on Unsupervised Agent Discovery (UAD) and the Bitwise Intelligence (B-IQ) metric, we formalize *preference-conditioned* prediction and control and derive a companion measure $\text{B-IQ}^{\text{want}}$. We show how to infer wants as prior preferences over observations in an active-inference lens, then evaluate systems via alignment ratios r_{pred} and r_{ctrl} that compare value-conditioned to raw channels. We argue that natural selection drives these ratios high in biological agents, while contemporary LLM training leaves them low. We propose training and evaluation protocols for LLM agent loops that directly raise $I_{\text{pred}}^{\text{want}}$ and $I_{\text{ctrl}}^{\text{want}}$, and we connect to related work in Inverse RL, RLHF/DPO, control-as-inference, world-model RL, and active inference.¹

1 Introduction

Distinguishing *predicting* P from *preferring* P is central to alignment discussions. The UAD framework discovers agents as Markov-blanket partitions in raw dynamics and exposes separate information flows for prediction and control [Zarncke, 2025a]. B-IQ adds a task-agnostic competence functional combining predictive and control mutual information with cost terms [Zarncke, 2025b]. Yet both quantities are *raw* channels: they measure total prediction and control, not necessarily directed at the agent’s goals.

This paper introduces a *preference-conditioned* refinement of these channels and shows how, for LLM agent loops, we can (i) infer “wants

” as prior preferences over future observations and (ii) measure competence at *realizing* those wants. The result sharpens why LLMs, though strong at I_{pred} , often underperform at $I_{\text{pred}}^{\text{want}}$ and $I_{\text{ctrl}}^{\text{want}}$ compared to biological agents.

Contributions

1. Define preference-conditioned channels $I_{\text{pred}}^{\text{want}}$ and $I_{\text{ctrl}}^{\text{want}}$ and the competence score $\text{B-IQ}^{\text{want}}$.
2. Derive inverse-active-inference estimators that recover wants as linear-feature priors over observations from LLM agent trajectories.
3. Propose alignment ratios $r_{\text{pred}}, r_{\text{ctrl}}$ and an evaluation protocol for humans vs. LLM agents.

¹Information quantities are in nats unless noted.

4. Discuss training interventions that raise the preference-conditioned channels and relate to CIRL, RLHF/DPO, ReAct/Toolformer/WebGPT, SayCan/PaLM-E, Dreamer, control-as-inference, and active inference.

2 Background: UAD and B-IQ

UAD. Given raw variables $\{X_k(t)\}$, UAD clusters an agent C when the blanket test holds $I(I_{t+1}^C; E_{t+1}^C | S_t^C, A_t^C) \approx 0$, then localizes memory by lagged mutual information and infers goals via an information-theoretic Lagrangian [Zarncke, 2025a]. UAD also formalizes inter-agent coupling and a cooperativity index that generalizes Hamilton’s rule [Zarncke, 2025a,c].

B-IQ. The baseline competence metric is

$$\text{B-IQ} = I_{\text{pred}} + I_{\text{ctrl}} - \beta H(I) - \gamma S, \quad (1)$$

with $I_{\text{pred}} = I(I_t; S_{t+1})$, $I_{\text{ctrl}} = I(A_t; E_{t+1})$, memory cost $H(I)$, and residual surprise $S = \mathbb{E}[-\log P(S_{t+1} | I_t)]$ [Zarncke, 2025b]. Physical bounds follow from sensor/actuator channel capacities.

3 Preference-Conditioned Channels

Let an agent’s “wants” be prior preferences over next observations, encoded as

$$\log P_{\text{pref}}(S') \propto \psi^\top \phi(S'), \quad (2)$$

with features $\phi(\cdot)$ (e.g., pass/fail, constraint violations, reward proxies, human-approval signals) and weights ψ to be inferred.

Value-conditioned prediction projects the predictive channel onto want-relevant sensory subspaces:

$$I_{\text{pred}}^{\text{want}} = I(I_t; S'[\psi]), \quad (3)$$

where $S'[\psi]$ indexes the sufficient statistics of $\phi(S')$.

Value-conditioned control similarly restricts control to the preferred externals:

$$I_{\text{ctrl}}^{\text{want}} = I(A_t; E'[\psi]). \quad (4)$$

Preference-conditioned competence is then

$$\text{B-IQ}^{\text{want}} = I_{\text{pred}}^{\text{want}} + I_{\text{ctrl}}^{\text{want}} - \beta H(I) - \gamma S, \quad (5)$$

with *alignment ratios*

$$r_{\text{pred}} := \frac{I_{\text{pred}}^{\text{want}}}{I_{\text{pred}}}, \quad r_{\text{ctrl}} := \frac{I_{\text{ctrl}}^{\text{want}}}{I_{\text{ctrl}}}. \quad (6)$$